



Mathematical
Institute

Kernel Limit of Recurrent Neural Networks Trained on Ergodic Data Sequences

Samuel Chun Hei Lam
joint work with Justin Sirignano and Konstantinos Spiliopoulos.

AI Security Reading Group

August 28, 2026

Oxford
Mathematics

Outline

1. Learning with time series
2. Fixed point method
3. Neural Tangent Kernel Analysis (ODE Method)

Paper available at [arXiv:2308.14555](https://arxiv.org/abs/2308.14555)

Learning with time series

Data sample need not be iid

- ▶ Input sequence $X := (X_k)_{k \geq 0} \subseteq \mathbb{R}^d$, could be “lifted” to a (geometric) *ergodic* Markov chain $(X_k, Z_k)_{k \geq 1} \subseteq \mathbb{R}^{d+1}$ that satisfies

$$(X_{k+1}, Z_{k+1}) \sim g(X_k, Z_k) + \epsilon_k$$

$(Z_k)_{k \geq 0}$ is *hidden*.

- ▶ Output sequence $Y := (Y_k)$ satisfies

$$Y_k = f(X_k, Z_k) + \eta_k.$$

- ▶ f, g are unknown.

Learning with time series

Objectives

- ▶ Want to learn map F such that $y_k \approx [F(X)]_k$ that minimises the empirical loss

$$L_T = \frac{1}{T} \sum_{k=1}^T \ell_k, \quad \ell_k = |Y_k - [F(X)]_k|^2. \quad (1)$$

Learning with time series

Objectives

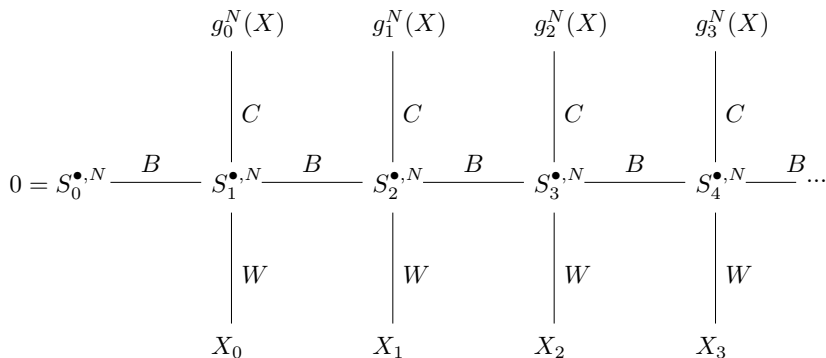
- ▶ Want to learn map F such that $y_k \approx [F(X)]_k$ that minimises the empirical loss

$$L_T = \frac{1}{T} \sum_{k=1}^T \ell_k, \quad \ell_k = |Y_k - [F(X)]_k|^2. \quad (1)$$

- ▶ Cannot build a fully-connected network that directly maps sequence x to y .
- ▶ *Recurrent* Neural Network (RNN): requires less parameters.
- ▶ Main applications: NLP (translation of words) - RNNs are popular before GPTs :(

Recurrent Neural Network (Elman '90)

Schematic Diagrams



Recurrent Neural Network (Elman '90)

The simplest architecture

The neural network is parametrised by weight vector/matrices $\theta = (C, W, B)$, with $C \in \mathbb{R}^N$, $W \in \mathbb{R}^{N \times d}$ and $B \in \mathbb{R}^N$.

$$S_{k+1}^{i,N}(x, \theta) = \sigma \left((W^i)^\top x_k + \frac{1}{N} \sum_{j=1}^N B^j S_k^{j,N}(x, \theta) \right)$$
$$g_k^N(x, \theta) = \frac{1}{N^\beta} \sum_{i=1}^N C^i S_{k+1}^{i,N}(x, \theta), \quad \beta \in (1/2, 1)$$

Training of Recurrent Neural Network

Backpropagation through time

- ▶ We can try to perform gradient descent to train the parameters:

$$\theta_{n+1} = \theta_n - \frac{\alpha}{N^{2-2\beta}} \nabla_{\theta} L_T$$

Training of Recurrent Neural Network

Backpropagation through time

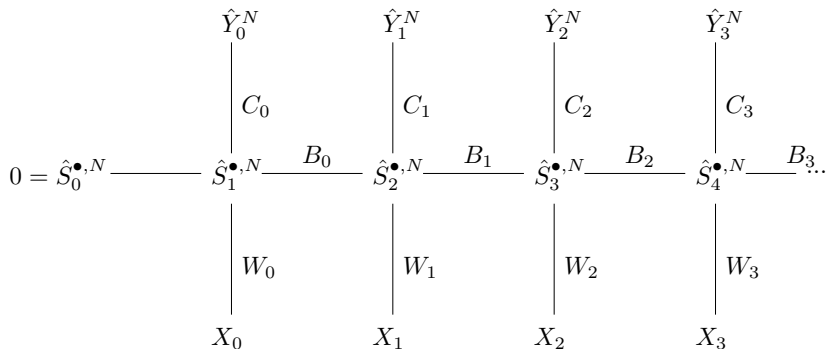
- ▶ We can try to perform gradient descent to train the parameters:

$$\theta_{n+1} = \theta_n - \frac{\alpha}{N^{2-2\beta}} \nabla_{\theta} L_T$$

- ▶ There are $O(T)$ term in the gradient $\nabla_{\theta} L_T$, so become intractable for large T .
- ▶ We want to learn the parameters “online”, i.e. update when we make a new observation (X_k, Y_k) .

Training of Recurrent Neural Network

Online Learning



$$\theta_{k+1} = \theta_k - \frac{\alpha}{N^{2-2\beta}} \text{“}\nabla_{\theta} \ell_k\text{”}$$

Training of Recurrent Neural Network (Elman '90)

Backpropagation through time

- ▶ Estimate the gradient $\nabla_{\theta} \ell_k$ by truncating the “backpropagation” by keeping the first (few) terms of the gradient.
- ▶ Does $(\theta_k)_{k \geq 1}$ converge? If so, does $(\theta_k)_{k \geq 1}$ converge to the correct minimum?

The beginning of Asymptotic Analysis

Two Components

- Define recursively

$$v_{k+1}^N(w) = \sigma \left(w^\top X_k + \frac{1}{N} \sum_{j=1}^N B^j v_k^N(w_{k-1}^i) \right), \quad v_0^N(w) \equiv 0.$$

- Parameters shall be encapsulated by $g_k^N : \mathcal{H} \rightarrow \mathbb{R}$, such that

$$g_k^N(h) := \frac{1}{N^\beta} \sum_{i=1}^N c_k^i h(w_k^i),$$

$$\mathcal{H} = \{\sigma(w^\top a + b) : a \in \mathbb{R}^d, |a| \leq 1, b \in \mathbb{R}\} \text{ with } H^1 \text{ norm.}$$

- Notice $\hat{Y}_k^N = g_k^N(v_{k+1}^N)$.

The beginning of Asymptotic Analysis

Two steps

$$\hat{Y}_k^N = g_k^N(v_{k+1}^N).$$

- ▶ We shall “collapse” the (distribution) of sequence of functions $(v_k^N)_{k \geq 0}$ into a single point.
- ▶ We shall study the limit of $g_{\lfloor Nt \rfloor}^N(\cdot)$ for $t \in [0, T]$ using the ODE method.

Recap: Neural Tangent Kernel Analysis (ODE Method)

- ▶ Given data $(x_j, y_j)_{j=1}^M$ such that $x_j = f(x_j)$ (+ noise).
- ▶ Guess f by

$$\hat{f}_{\theta, \beta}(x) = \frac{1}{N^\beta} \sum_{i=1}^N c^i \sigma((w^i)^\top x), \quad \theta = (C, W),$$

- ▶ Say want to minimise

$$L(\theta; \beta) = \frac{1}{M} \sum_{j=1}^M |y_j - \hat{f}_{\theta, \beta}(x_j)|^2. \quad (2)$$

using (stochastic) gradient descent

$$\theta_{k+1} = \theta_k - \frac{\alpha}{N^{2-2\beta}} \nabla_{\theta} |y_{j_k} - \hat{f}_{\theta_k, \beta}(x_{j_k})|^2$$

Recap: Neural Tangent Kernel Analysis (ODE Method)

- The gradient updates are the following

$$\begin{aligned}c_{k+1}^i - c_k^i &= 2\alpha \left[\left(y_{j_k} - \hat{f}_{\theta_k}(x_{j_k}) \right) \frac{1}{N^{2-\beta}} \sigma \left((w_k^i)^\top x_{j_k} \right) \right] \\w_{k+1}^i - w_k^i &= 2\alpha \left[\left(y_{j_k} - \hat{f}_{\theta_k}(x_{j_k}) \right) \frac{1}{N^{2-\beta}} c_k^i \sigma' \left((w_k^i)^\top x_{j_k} \right) x_{j_k} \right]\end{aligned}$$

so summing yields the following evolution (ignoring higher-order term)

$$\begin{aligned}\hat{f}_{\theta_{k+1},\beta}(x) - \hat{f}_{\theta_k,\beta}(x) &\approx \frac{2\alpha}{N} (y_{j_k} - \hat{f}_{\theta_k}(x_{j_k})) \\&\times \frac{1}{N} \sum_{i=1}^N \left[\sigma \left((w_k^i)^\top x_{j_k} \right) \sigma \left((w_k^i)^\top x \right) \right. \\&\quad \left. + c_k^2 \sigma' \left((w_k^i)^\top x_{j_k} \right) \sigma' \left((w_k^i)^\top x \right) x_{j_k}^\top x \right]\end{aligned}$$

Recap: Neural Tangent Kernel Analysis (ODE Method)

- Let $A_{x,x',k}^N$ be the symmetric kernel (matrix)

$$A_{x,x',k}^N = \frac{1}{N} \sum_{i=1}^N \left[\sigma \left((w_k^i)^\top x \right) \sigma \left((w_k^i)^\top x' \right) + (c_k^i)^2 \sigma' \left((w_k^i)^\top x \right) \sigma' \left((w_k^i)^\top x' \right) x^\top x' \right],$$

then

$$\hat{f}_{\theta_{\lfloor Nt \rfloor}, \beta}(x) - \hat{f}_{\theta_0, \beta}(x) \approx \frac{1}{N} \sum_{k=0}^{\lfloor Nt \rfloor - 1} 2\alpha(y_{j_k} - \hat{f}_{\theta_k}(x_{j_k})) A_{x, x_{j_k}, k}^N.$$

Recap: Neural Tangent Kernel Analysis (ODE Method)

- ▶ Assuming initial parameters $\theta_0 = (c_0^i, w_0^i)_{i=1}^N$ are i.i.d. randomly initialised with distribution λ , $c_0^i \perp w_0^i$, then for $k \leq NT$, the parameters do not move too far away from initialisation on average:

$$\mathbb{E}|c_k^i - c_0^i|^2 \vee \mathbb{E}|w_k^i - w_0^i|^2 = o_N(1).$$

Recap: Neural Tangent Kernel Analysis (ODE Method)

- Assuming initial parameters $\theta_0 = (c_0^i, w_0^i)_{i=1}^N$ are i.i.d. randomly initialised with distribution λ , $c_0^i \perp w_0^i$, then for $k \leq NT$, the parameters do not move too far away from initialisation on average:

$$\mathbb{E}|c_k^i - c_0^i|^2 \vee \mathbb{E}|w_k^i - w_0^i|^2 = o_N(1).$$

- Also by “law of large numbers”,

$$\begin{aligned} A_{x,x',0}^N &\approx A_{x,x'}^\infty := \mathbb{E}_{(c,w) \sim \lambda} \left[\sigma(w^\top x) \sigma(w^\top x') \right. \\ &\quad \left. + c^2 \sigma'(w^\top x) \sigma'(w^\top x') x^\top x' \right], \end{aligned}$$

Recap: Neural Tangent Kernel Analysis (ODE Method)

- With (x_{j_k}, y_{j_k}) are sampled randomly uniform, we shall have

$$\hat{f}_{\theta_{\lfloor Nt \rfloor}, \beta}(x) - \hat{f}_{\theta_0, \beta}(x) \approx \frac{1}{N} \sum_{k=0}^{\lfloor Nt \rfloor - 1} \frac{2\alpha}{M} \sum_{j=1}^M (y_j - \hat{f}_{\theta_k}(x_j)) A_{x, x_j}^{\infty}.$$

Recap: Neural Tangent Kernel Analysis (ODE Method)

- ▶ With (x_{j_k}, y_{j_k}) are sampled randomly uniform, we shall have

$$\hat{f}_{\theta_{\lfloor Nt \rfloor}, \beta}(x) - \hat{f}_{\theta_0, \beta}(x) \approx \frac{1}{N} \sum_{k=0}^{\lfloor Nt \rfloor - 1} \frac{2\alpha}{M} \sum_{j=1}^M (y_j - \hat{f}_{\theta_k}(x_j)) A_{x, x_j}^{\infty}.$$

- ▶ This is the discretisation of the following gradient flow **ODE**:

$$\frac{d\hat{f}_{\theta_t, \beta}(x)}{dt} = \frac{2\alpha}{M} \sum_{j=1}^M (y_j - \hat{f}_{\theta_t}(x_j)) A_{x, x_j}^{\infty}.$$

- ▶ We only need to solve for $\hat{\mathbf{f}}_t := \left(\hat{f}_{\theta_t, \beta}(x_j) \right)_{j=1}^M$ (without accessing actual θ_t).

Recap: Neural Tangent Kernel Analysis (ODE Method)

- If we let $L_t = \|\mathbf{y} - \hat{\mathbf{f}}_t\|^2$, with $\mathbf{y} = (y_j)_{j=1}^M$, we shall have

$$\frac{dL_t}{dt} = -\frac{4\alpha}{M^2}(\mathbf{y} - \hat{\mathbf{f}}_t)^\top A^\infty(\mathbf{y} - \hat{\mathbf{f}}_t).$$

- If we know that the kernel A^∞ has $\lambda_{\min} > 0$, then

$$\frac{dL_t}{dt} \leq -\frac{4\alpha}{M^2}\lambda_{\min}L_t,$$

so $L_t \rightarrow 0$ exponentially fast!

Fixed point analysis

Exact computation of gradient (Almeida '87, Pineda '87)

- ▶ Let's say the input sequence is $x = (x, x, \dots)$, and $|\sigma'| \leq C_\sigma < 1$.
- ▶ One could select (W^i, B^i) at *fixed point*, such that

$$S_{k+1}^{i,N}(x, \theta) = S_k^{i,N}(x, \theta) = s^{i,N}(x, \theta),$$

such that

$$s^{i,N}(x, \theta) = \sigma \left((w^i)^\top x + \frac{1}{N} \sum_{j=1}^N B^j s^{j,N}(x, \theta) \right).$$

Fixed point analysis

Exact computation of gradient (Almeida '87, Pineda '87)

- ▶ Let's say the input sequence is $x = (x, x, \dots)$, and $|\sigma'| \leq C_\sigma < 1$.
- ▶ One could select (W^i, B^i) at *fixed point*, such that

$$S_{k+1}^{i,N}(x, \theta) = S_k^{i,N}(x, \theta) = s^{i,N}(x, \theta),$$

such that

$$s^{i,N}(x, \theta) = \sigma \left((w^i)^\top x + \frac{1}{N} \sum_{j=1}^N B^j s^{j,N}(x, \theta) \right).$$

$\Rightarrow$ Exact computation of the gradients ∇L_T .

Fixed point analysis

Contribution: random fixed point

$$(X_{k+1}, Z_{k+1}) \sim g(X_k, Z_k) + \epsilon_k$$

- ▶ What if we know that $(X_k)_{k \geq 1}$ is geometric ergodic (i.e. *converges* in distribution exponentially fast)?
- ▶ View $S_k^{i,N} = h_k^N(W^i)$, so that

$$h_{k+1}^N(w) = \sigma \left(w^\top X_k + \frac{1}{N} \sum_{j=1}^N B^j h_k^N(W^j) \right).$$

- ▶ Assume (C^i, W^i, B^i) iid $\sim \lambda$ and are independent of (X_k, Z_k) .

Fixed point analysis

Contribution: random fixed point

$$\begin{aligned}h_{k+1}^N(w) &= \sigma \left(w^\top X_k + \frac{1}{N} \sum_{j=1}^N B^j h_k^N(W^j) \right), \\h_{k+1}(w) &= \sigma \left(w^\top X_k + \mu_b \int h_k(w) \lambda(dw) \right), \quad \mu_b = \mathbb{E}[B^j], \\ \|h\|_{H^1}^2 &= \int (|h|^2 + |\nabla h|^2) \lambda(dw).\end{aligned}$$

Lemma. There exists $C, C_T > 0$ such that

$$\begin{aligned}\forall k \leq NT, \quad \mathbb{E} \|v_k^N - h_k^N\|_{H^1}^2 &= o_N(1), \\ \forall k \geq 0, \quad \mathbb{E} \|h_k^N - h_k\|_{H^1}^2 &\leq C/N.\end{aligned}$$

Fixed point analysis

Contribution: random fixed point

$$\begin{aligned} h_{k+1}(w) &= \sigma \left(w^\top X_k + \mu_b \int h_k(w) \lambda(dw) \right) \\ &= \sigma \left(w^\top X_k + \mu_b \langle h_k(w), \lambda \rangle \right), \quad \mu_b = \mathbb{E}[B^i] \end{aligned}$$

- ▶ $H_k = (X_k, Z_k, h_k)$ is a time-homogeneous Markov chain.
- ▶ This has *contractive* transition operator $P(h, A)$, such that

$$\frac{\text{Wass}_2(P(H, \cdot), P(H', \cdot))}{\|H - H'\|} \leq q < 1,$$

i.e. exists coupling $\gamma_{H, H'}$ such that

$$\int \|h - h'\|^2 \gamma_{H, H'}(dh, dh') \leq q^2 \|H - H'\|^2$$

Fixed point analysis

Contribution: random fixed point

► **Theorem.** this Markov chain has stationary distribution μ

Fixed point analysis

Contribution: random fixed point

- ▶ **Theorem.** this Markov chain has stationary distribution μ
 $\implies$ this is the “random fixed point”.

Fixed point analysis

Contribution: random fixed point

- ▶ **Theorem.** this Markov chain has stationary distribution μ
 $\implies$ this is the “random fixed point”.
- ▶ **Remark.** if concerned with defining integral over function spaces, then one could look at the joint process (X_k, Z_k, u_k) instead, where $u_k = \mu_b \int h_k(w) \lambda(dw)$.

Converging to limiting ODE

Deriving the ODE

- Recall the functional

$$g_k^N(h) := \frac{1}{N^\beta} \sum_{i=1}^N c_k^i h(w_k^i).$$

Converging to limiting ODE

Deriving the ODE

- Recall the functional

$$g_k^N(h) := \frac{1}{N^\beta} \sum_{i=1}^N C_k^i h(W_k^i).$$

- Choosing $\beta \in (1/2, 1)$, one could Taylor expand to look at the difference $\Delta g_k^N(h) := g_{k+1}^N(h) - g_k^N(h)$.

Converging to limiting ODE

Deriving the ODE

► Let

$$\lambda_k^N = \sum_{i=1}^N \delta_{(C_k^i, W_k^i, B_k^i)}, \quad \langle f, \lambda \rangle = \int f d\mu.$$

► Then,

$$\begin{aligned} \Delta g_k^N(h) &\approx -\frac{\alpha}{N} (\psi^N(g_k^N(v_{k+1}^N)) - Y_k) (\langle v_{k+1}^N(w) h(w) \\ &\quad + c^2 \sigma' \left(w^\top X_k + \langle b v_k^N(w), \lambda_k^N \rangle \right) \nabla h(w)^\top X_k, \lambda_k^N \rangle), \\ v_{k+1}^N(w) &= \sigma \left(w^\top X_k + \frac{1}{N} \sum_{j=1}^N B_k^j v_k^N(W_{k-1}^j) \right). \end{aligned}$$

Converging to limiting ODE

Deriving the ODE

► Let

$$\lambda_k^N = \sum_{i=1}^N \delta_{(C_k^i, W_k^i, B_k^i)}, \quad \langle f, \lambda \rangle = \int f d\mu.$$

► The parameters does not move too far way from initialisation for $k \leq NT$, so:

$$\begin{aligned} \Delta g_k^N(h) &\approx -\frac{\alpha}{N} (\psi^N(g_k^N(h_{k+1}^N)) - Y_k) (\langle h_{k+1}^N(w) h(w) \\ &\quad + c^2 \sigma' \left(w^\top X_k + \langle b h_k^N(w), \lambda_0^N \rangle \right) \nabla h(w)^\top X_k, \lambda_0^N \rangle), \end{aligned}$$

$$h_{k+1}^N(w) = \sigma \left(w^\top X_k + \frac{1}{N} \sum_{j=1}^N B_0^j h_k^N(W_0^j) \right).$$

Converging to limiting ODE

Deriving the ODE

- ▶ We look at the functional

$$g_k^N(h) := \frac{1}{N^{\beta_2}} \sum_{i=1}^N C_k^i h(W_k^i), \quad h \in H^1.$$

- ▶ By weak law of large number

$$\begin{aligned} \Delta g_k^N(h) &\approx -\frac{\alpha}{N} (\psi^N(g_k^N(h_{k+1})) - Y_k) (\langle h_{k+1}(w) h(w) \\ &\quad + c^2 \sigma' \left(w^\top X_k + \langle b h_k(w), \lambda \rangle \right) \nabla h(w)^\top X_k, \lambda \rangle), \\ h_{k+1}(w) &= \sigma \left(w^\top X_k + \mu_b \int h_k(w) \lambda(dw) \right). \end{aligned}$$

Converging to limiting ODE

Deriving the ODE

- ▶ We look at the functional

$$g_k^N(h) := \frac{1}{N^{\beta_2}} \sum_{i=1}^N C_k^i h(W_k^i), \quad h \in H^1.$$

- ▶ By weak law of large number

$$\begin{aligned} \Delta g_k^N(h) &\approx -\frac{\alpha}{N} (\psi^N(g_k^N(h_{k+1})) - Y_k) (\langle h_{k+1}(w) h(w) \\ &\quad + c^2 \sigma' \left(w^\top X_k + \langle b h_k(w), \lambda \rangle \right) \nabla h(w)^\top X_k, \lambda \rangle), \\ h_{k+1}(w) &= \sigma \left(w^\top X_k + \mu_b \int h_k(w) \lambda(dw) \right). \end{aligned}$$

- ▶ One could further replace (X_k, h_k) with their averages (or the random fixed point) using a Poisson equation argument.

Converging to limiting ODE

Deriving the ODE

- ▶ We look at the functional

$$g_k^N(h) := \frac{1}{N^{\beta_2}} \sum_{i=1}^N C_k^i h(W_k^i), \quad h \in H^1.$$

- ▶ By weak law of large number

$$\begin{aligned} \Delta g_k^N(h) &\approx -\frac{\alpha}{N} (\psi^N(g_k^N(h_{k+1})) - Y_k) (\langle h_{k+1}(w) h(w) \\ &\quad + c^2 \sigma' \left(w^\top X_k + \langle b h_k(w), \lambda \rangle \right) \nabla h(w)^\top X_k, \lambda \rangle), \\ h_{k+1}(w) &= \sigma \left(w^\top X_k + \mu_b \int h_k(w) \lambda(dw) \right). \end{aligned}$$

- ▶ One could further replace (X_k, h_k) with their averages (or the random fixed point) using a Poisson equation argument.

Poisson Equation

Relating estimates with averages

- Given G a measurable function with $\int |G| d\mu < +\infty$, the Poisson equation is a functional equation on $\hat{G}$

$$\hat{G}(H) - \int \hat{G}(\bar{H}) P(H, d\bar{H}) = G(H) - \int G(\bar{H}) \mu(d\bar{H}),$$

where $H = (x, z, y, h)$, $\bar{H} = (\bar{x}, \bar{z}, \bar{y}, \bar{h})$, and P is the transitional operator of the Markov chain.

Poisson Equation

Relating estimates with averages

- ▶ Given G a measurable function with $\int |G| d\mu < +\infty$, the Poisson equation is a functional equation on $\hat{G}$

$$\hat{G}(H) - \int \hat{G}(\bar{H}) P(H, d\bar{H}) = G(H) - \int G(\bar{H}) \mu(d\bar{H}),$$

where $H = (x, z, y, h)$, $\bar{H} = (\bar{x}, \bar{z}, \bar{y}, \bar{h})$, and P is the transitional operator of the Markov chain.

- ▶ A solution would be

$$\hat{G}(H) = \sum_{k=0}^{\infty} \left(\int G(\bar{H}) P^k(H, d\bar{H}) - \int G(\bar{H}) \mu(d\bar{H}) \right),$$

Poisson Equation

Relating estimates with averages

► Let $G^{N,k}(H_k, \nu) := \Delta g_k^N(\nu)$, and

$$\begin{aligned} \Delta \tilde{g}_k^N(\nu) &= -\frac{\alpha}{N} \int (g_k^N(\varsigma(h)) - y) (\langle \varsigma(h) \nu(w) \\ &\quad + c^2 \sigma' \left(w^\top x + \langle bh(w), \lambda \rangle \right) \nabla v(w)^\top x, \lambda \rangle) d\mu, \\ [\varsigma(h)](w) &= \sigma \left(w^\top X_k + \mu_b \int h_k(w) \lambda(dw) \right). \end{aligned}$$

► Then

$$\begin{aligned} \Delta g_k^N(\nu) - \Delta \tilde{g}_k^N(\nu) &= G_k^N(H_k, \nu) - \int G_k^N(\bar{H}, \nu) \mu(d\bar{H}) \\ &= \hat{G}_k^N(H_k, \nu) - \int \hat{G}_k^N(\bar{H}, \nu) P(H_k, d\bar{H}) \end{aligned}$$

Poisson Equation

Relating estimates with averages

- ▶ One can then show that $\Delta g_k^N(v) \approx \Delta \tilde{g}_k^N(v)$, so

$$g_{\lfloor Nt \rfloor}^N(v) \approx \sum_{k=0}^{\lfloor Nt \rfloor - 1} \int G_k^N(\bar{H}, v) \mu(d\bar{H}).$$

This is the discretisation of the following ODE -

Converging to limiting ODE

Deriving the ODE

Theorem. Consider the infinite-dimensional ODE

$$g_t(h) = -\alpha \int_0^t \left[\int_{\mathcal{X}} (g_s(\varsigma(H)) - y) \mathcal{K}_{x,\lambda}(h, h) \mu(dx, dz, dy, dh) \right] ds,$$

$$g_0(h) = 0,$$

$$\mathcal{K}_{x,\lambda}(h, h) = \langle \sigma(w^\top x + \langle b'h(w'), \lambda \rangle) h(w) \\ + c^2 \sigma'(w^\top x + \langle b'h(w'), \lambda \rangle) \nabla h(w)^\top x, \lambda \rangle,$$

$$[\varsigma(H)](w) = \sigma(w^\top x + \langle b'h(w'), \lambda \rangle)$$

Then

$$\sup_{h \in \mathcal{H}} \sup_{t \in [0, T]} \mathbb{E} |g_{\lfloor Nt \rfloor}^N(h) - g_t(h)| = o_N(1). \quad (3)$$

Open questions

Corollary. With the limit ODE, we can show that

$$\int |y - g_t(h)|^2 \mu(dx, dz, dy, dh) \quad (4)$$

is decreasing along t , where μ is the *random fixed point*.

Open questions

Corollary. With the limit ODE, we can show that

$$\int |y - g_t(h)|^2 \mu(dx, dz, dy, dh) \quad (4)$$

is decreasing along t , where μ is the *random fixed point*. **Can we do better?**

Extension.

- ▶ What happen if we have different choice of scalings β ?
- ▶ Other RNN architectures: LSTM, xLSTM, Llama...

Open questions

Corollary. With the limit ODE, we can show that

$$\int |y - g_t(h)|^2 \mu(dx, dz, dy, dh) \quad (4)$$

is decreasing along t , where μ is the *random fixed point*. **Can we do better?**

Extension.

- ▶ What happen if we have different choice of scalings β ?
- ▶ Other RNN architectures: LSTM, xLSTM, Llama...

Thank you.